

UNIT-IV

Latin Square Design: According to Column, Row & Varieties are arranged in square is known as Latin Square. It is very

We consider an (agricultural experiment) Population in which n^2 plots are taken and arranged in the form of an $n \times n$ square, such that the plots in each row will be homogeneous as far as possible with respect to one factor of classification, say, soil fertility and plots in each column will be homogeneous as far as possible with respect to another factor of classification, say, seed quality.

Then n treatments are given to these plots such that each treatment occurs only once in each row and only once in each column. The various possible arrangements obtained in this manner are known as Latin squares of order n . This design of experiment is called the Latin Square Design (L.S.D)

Analysis of Variance for Three Factors of classification

- 1) $N = n^2 = (\text{order of square})^2$
- 2) Grand total G
- 3) Correction Factor, $C.F = \frac{G^2}{N}$
- 4) The sum of squares = $\sum y_{ijk}^2 - C.F$
- 5) SS between Rows = $\frac{\sum R_i^2}{n} - C.F$
- 6) SS between Columns = $\frac{\sum C_j^2}{n} - C.F$
- 7) SS between treatment = $\frac{\sum T_k^2}{n} - C.F$

8) Error sum of squares (should always > 0)
 = Total SS - SS btw Row - SS btw column
 - SS btw treatment

ANOVA for L.S.D (Latin Square Design)

Sources of Variation S.V	Degrees of freedom D.F	Sum of Squares S.S	Mean sum of squares $M.S = \frac{S.S}{D.F}$	F_s	F table Value $F_{(r, E)}$
Between rows	$r-1$	SS_R	MS_R	$\frac{M.S_R}{MSE}$	$F_{(r-1, E)}$
Between columns	$C-1$	SS_C	MS_C	$\frac{MS_C}{MSE}$	$F_{(C-1, E)}$
Between Treatment	$T-1$	SST	MST	$\frac{MST}{MSE}$	$F_{(T-1, E)}$
Error	E	SS_E	MSE		
Total	$N-1$				

Inference:

Rows	Columns	Treatment
$F_o = \text{observed Value}$	F_o	F_o
$F_e = \text{table Value}$	F_e	F_e
If $F_o < F_e$, We infer that there is no significant difference between the rows.	If $F_o < F_e$, We infer that there is no significant difference between the rows columns	Suppose $F_o > F_e$, We infer that there is a significant difference between the treatments.

Example for LSD:

A	B
B	A

2x2

A	B	C
B	C	A
C	A	B

3x3

A	B	C	D
B	C	D	A
C	D	A	B
D	A	B	C

4x4

Design of Experiments

Theory	Application
<u>Analysis of Variance</u>	<u>Design of Experiments</u>
One-Way	Single factor (E.R.D) Completely Randomized Design
Two-Way	Two factors (R.B.D) Randomized Block Design
Three-Way	Three factors (L.S.D) Latin Square Design.

Single Factor Effect:

Effect of drugs for an ailment (Headache)

"Variations in Recovery time" - Completely Randomized Design.

Drugs			
D ₁	D ₂	D ₃	D ₄
5	7	10	4
7	8	5	6
6	10	7	5
10	6	8	11

Two Factor Effect:

Effect of age and drugs.

"Variations in Recovery time & drugs" - Randomized Block Design

Age	Drugs			
	D ₁	D ₂	D ₃	D ₄
<14	10	13	15	12
14-35	8	10	12	15
35-60	7	10	14	15
60+	20	19	22	25

Three Factor Effect:

Effect of drug, Age and Occupation:

(Occupation: A-Teacher, B-Office Manager,
C-Head, D-Office Assistant)

"Variations w.r.to Recovery time, occupation
and drugs" - Latin Square Design.

Age	Drugs			
	D ₁	D ₂	D ₃	D ₄
I	B 14 min	C 12 min	D 10 min	A 7 min
II	D 13 min	A 15 min	B 8 min	C 11 min
III	C 16 min	B 12 min	A 10 min	D 5 min
IV	A 4 min	D 7 min	C 8 min	B 6 min

Drugs			
D ₁	D ₂	D ₃	D ₄
10	7	8	10
10	5	10	10
10	7	10	10
11	8	10	10

- 1) The following data is related to the measurement of cholesterol content (%) performed in 4 different laboratories w.r.to 4 diet foods. The design adopted is a Randomized Block design.

Diet Food	Laboratory				Total
	I	II	III	IV	
A	4	5	6	4	19
B	3	4	5	8	20
C	5	3	7	5	20
D	2	3	4	2	11
Total	14	15	22	19	70

Analyze the data and comment on your result.
 [Given $F_{0.05} = 3.86$ for (8,9) d.f.]

Sol:

Effects of Laboratory and Diet Food - RBD

Rows, $R = 4$; Columns, $C = 4$

1) Total no. of observations,

$$N = R \times C = 4 \times 4 = 16$$

2) Grand total, $G = \sum_i R_i + \sum_j C_j = 70$

3) Correction Factor, $C.F = \frac{G^2}{N} = \frac{70^2}{16} = 306.25$

4) Total sum of squares, $SS = \sum_{ij} y_{ij}^2 - C.F$
 $= 348 - 306.25 = 41.75$

5) SS btw Rows, $SSR = \frac{\sum R_i^2}{C} - C.F$
 $= \frac{19^2 + 20^2 + 20^2 + 11^2}{4} - 306.25$
 $= 320.5 - 306.25$
 $= 14.25$

$$b) \text{ SS btw columns, } SSC = \frac{\sum C_j^2}{R} - CF$$

$$= \frac{14^2 + 15^2 + 22^2 + 19^2}{4} - 1225$$

$$= 316.5 - 1225 = 306.25$$

$$= 10.25$$

$$7) \text{ Error SS} = \text{Total SS} - \text{SSR} - \text{SSC}$$

$$(Should > 0) = 41.75 - 14.25 - 10.25$$

$$= 17.25 > 0$$

Analysis:

ANOVA

Sources of Variation	D.F	SS	MS = $\frac{SS}{D.F}$	$\frac{MS}{E} F_0$	F_e
Between Rows	$R-1=3$	14.25	$\frac{14.25}{3} = 4.75$	$\frac{4.75}{1.92} = 2.47$	3.86
Between columns	$C-1=3$	10.25	$\frac{10.25}{3} = 3.42$	$\frac{3.42}{1.92} = 1.78$	3.86
Error	9	17.25	$\frac{17.25}{9} = 1.92$		
Total	$N-1=15$				

Inference

Rows Diet Food	Columns Laboratory
$F_0 = 2.47$ $F_e = 3.86$	$F_0 = 1.78$ $F_e = 3.86$
Since $F_0 < F_e$, We infer that there is no significant difference in cholesterol % between Diet Foods.	Since $F_0 < F_e$, We infer that there is no significant difference in cholesterol % between Laboratories.

2) A company Manufacturing a consumer product appointed four Salesmen & observed their sales in 3 different states. The no. of products sold is given below

State	Salesmen				Total
	A	B	C	D	
I	22	27	38	45	132
II	28	32	40	35	135
III	25	40	36	22	123

The design adopted is a randomized block design. Analyse the data & comment on your result
 $F_{0.05}(3,6) = 4.76$, $F_{0.05}(2,6) = 5.14$

Sol:

Effects of Salesmen and state - RBD

State: Rows, $R = 3$

Salesmen: columns, $C = 4$

1) $N = R \times C = 3 \times 4 = 12$

2) Grand total, $G = \sum R_i = 780$

3) Correction Factor, $C.F = \frac{G^2}{N} = 12675$

4) Total sum of squares, $SS = \sum y_{ij}^2 - C.F$
 $= 13320 - 12675$
 $= 645$

5) SS btw Rows, $SSR = \frac{\sum R_i^2}{C} - C.F$
 $= \frac{132^2 + 135^2 + 123^2}{4} - 12675$
 $= 19.5$

$$\begin{aligned}
 \text{b) ss btw columns, } SSC &= \frac{\sum G^2}{R} - CF \\
 &= \frac{75^2 + 99^2 + 114^2 + 102^2}{3} - 12675 \\
 &= 267
 \end{aligned}$$

$$\begin{aligned}
 \text{7) Error ss} &= \text{Total ss} - SSR - SSC \\
 &= 645 - 19.5 - 267 \\
 &= 358.5 > 0
 \end{aligned}$$

Analysis:

ANOVA

Sources of Variations	DF	SS	MS	F_0	F_e
Between Rows	$R-1=2$	19.5	9.75	0.1632	5.14
Between columns	$C-1=3$	267	89	1.4895	4.76
Error	6	358.5	59.75		
Total	$N-1=11$				

Inference:

Rows (State)	Columns Salesmen
$F_0 = 0.1632$ $F_e = 5.14$ Since $F_0 < F_e$, We infer that there is no significant difference in no. of products manufactured between states.	$F_0 = 1.4895$ $F_e = 4.76$ Since $F_0 < F_e$, We infer that there is no significant difference in no. of products manufactured between Salesmen.

- 3) The following data is related to the relief time in minutes of the patients of various age groups after administering the 3 drugs for an ailment. Analyze the data & comment on your results [$F_{0.05}(2,6) = 5.14$; $F_{0.05}(3,6) = 4.76$]

Age of Patients	Drugs			Total
	A	B	C	
<14	4	3	2	9
14-35	2	2	1	5
35-60	3	1	0	4
60 ≥	1	2	4	7
Total	10	8	7	25

Sol:

Effects of Age and Drugs. - RBD

Age: Rows, $R = 4$

Drugs: columns, $C = 3$

- $N = R \times C = 4 \times 3 = 12$
- Grand total, $G = \sum R_i = 25$
- Correction Factor, $CF = \frac{G^2}{N} = 52.08$
- Total SS = $\sum y_{ij}^2 - CF$
 $= 69 - 52.08$
 $= 16.92$
- SS btw Rows, $SSR = \frac{\sum R_i^2}{C} - CF$
 $= \frac{9^2 + 5^2 + 4^2 + 7^2}{3} - 52.08$
 $= 4.92$
- SS btw columns, $SSC = \frac{\sum C_j^2}{R} - CF$
 $= \frac{10^2 + 8^2 + 7^2}{4} - 52.08$
 $= 1.17$
- Error SS = Total SS - SSR - SSC = $16.92 - 4.92 - 1.17 = 10.83$

Analysis:

ANOVA

Source of Variations	DF	SS	MS	F_0	F_e
Between Rows	$R-1=3$	4.92	1.64	0.1514	4.76
Between Columns	$C-1=2$	1.17	0.585	0.324	5.14
Error	6	10.83	1.805		
Total	$N-1=11$				

Inference:

Rows (Age)	Columns (Drugs)
$F_0 = 0.1514$ $F_e = 4.76$ Since $F_0 < F_e$, We infer that there is no significant difference in relief time btw age	$F_0 = 0.324$ $F_e = 5.14$ Since $F_0 < F_e$, We infer that there is no significant difference in relief time btw drugs.

- 4) Analyze the data given in the following Latin Square design and comment on your results. The figures in brackets are related to the yield (kg) of a crop. Given $F_{0.05} = 4.76$ for (3,6) d.f

A(5)	B(6)	C(10)	D(7)	28	Treatment
B(5)	C(10)	D(8)	A(6)	29	A=29
C(11)	D(9)	A(8)	B(7)	35	B=22
D(6)	A(10)	B(4)	C(9)	29	C=40
Total	27	35	30	29	D=30
				121	<u>121</u>

1) $N = n^2 = 4^2 = 16$

2) Grand total, $G = 121$

3) Correction Factor, $C.F = \frac{G^2}{N} = 915.06$

4) Total $SS = \sum y_{ijk}^2 - C.F$
 $= 983 - 915.06$
 $= 67.94$

5) SS btw Rows, $SSR = \frac{\sum R_i^2}{n} - C.F$
 $= \frac{28^2 + 29^2 + 35^2 + 29^2}{4} - 915.06$
 $= 7.69$

6) SS btw columns, $SSC = \frac{\sum C_j^2}{n} - C.F$
 $= \frac{27^2 + 35^2 + 30^2 + 29^2}{4} - 915.06$
 $= 8.69$

7) SS btw Treatments, $SST = \frac{\sum M_k^2}{n} - C.F$
 $= \frac{29^2 + 22^2 + 40^2 + 30^2}{4} - 915.06$
 $= 41.19$

$$\begin{aligned}
 8) \text{ Error SS} &= \text{Total SS} - \text{SSR} - \text{SSC} - \text{SST} \\
 &= 67.94 - 7.69 - 8.69 - 41.19 \\
 &= 10.37 > 0
 \end{aligned}$$

Analysis:

ANOVA					
Sources of Variations	Degrees of Freedom	Sum of Squares	Mean sum of squares	F_0	F_e
Between rows	$r-1 = 3$	7.69	2.56	$\frac{1.48}{2.1865}$	4.76
Between columns	$c-1 = 3$	8.69	2.8967	$\frac{1.67}{0.211}$	4.76
Between Treatments	$T-1 = 3$	41.19	13.73	7.94	4.76
Error	6	10.37	1.73		
Total	$N-1 = 15$				

Inference:

Rows	Columns	Treatments
$F_0 = 1.48$ $F_e = 4.76$ Since $F_0 < F_e$, We infer that there is no significant difference btw rows	$F_0 = 1.67$ $F_e = 4.76$ Since $F_0 < F_e$, We infer that there is no significant diff. btw Columns	$F_0 = 7.94$ $F_e = 4.76$ Since $F_0 > F_e$, We infer that there is significant diff. btw Treatments.

- 5) An oil company is interested in testing four ^{different} blends of gasoline for the fuel efficiency by controlling the variability of four different drivers and four models of cars. The fuel efficiency was measured as kilometer per litre. After driving the cars over a standard course. Data is presented in a 4x4 Latin Square Design.

A(13)	D(9)	C(10)	B(12)	44	A=53
B(12)	C(10)	A(13)	D(8)	43	B=48
C(10)	B(12)	D(9)	A(13)	44	C=40.5
D(9)	A(14)	B(12)	C(10.5)	45.5	D=35
Total				176.5	176.5
44	45	44	43.5		

Sol:

- 1) $N = n^2 = 4^2 = 16$
- 2) Grand total, $G = 176.5$
- 3) Correction Factor, $C.F = \frac{G^2}{N} = 1947.02$
- 4) Total SS = $\sum y_{ijk}^2 - C.F$
 $= 1996.25 - 1947.02 = 49.23$
- 5) SS btw Rows, $SSR = \frac{\sum R_i^2}{n} - C.F$
 $= \frac{44^2 + 43^2 + 44^2 + 45.5^2}{4} - 1947.02$
 $= 0.79$
- 6) SS btw columns, $SSC = \frac{\sum C_j^2}{n} - C.F$
 $= \frac{44^2 + 45^2 + 44^2 + 43.5^2}{4} - 1947.02$
 $= 0.29$

$$\begin{aligned}
 7) \text{ SS between Treatments, } SST &= \frac{\sum Y_k^2}{n} - CF \\
 &= \frac{53^2 + 48^2 + 40.5^2 + 35^2}{4} - 1947.02 \\
 &= 47.54
 \end{aligned}$$

$$\begin{aligned}
 8) \text{ Error sum of squares,} \\
 &= \text{Total SS} - \text{SSR} - \text{SSC} - \text{SST} \\
 &= 49.23 - 0.79 - 0.29 - 47.54 \\
 &= 0.61 > 0
 \end{aligned}$$

Analysis:

ANOVA For LSD

Sources of Variations	Degrees of Freedom	Sum of Squares	Mean sum of Squares	F_0	F_e
Btw rows	3	0.79	0.2633	2.59	4.76
Btw columns	3	0.29	0.0967	0.95	4.76
Btw Treatments	3	47.54	15.8467	155.82	4.76
Error	6	0.61	0.1017		
Total	15				

Inference:

Rows	Columns	Treatments
$F_0 = 2.59$ $F_e = 4.76$	$F_0 = 0.95$ $F_e = 4.76$	$F_0 = 155.82$ $F_e = 4.76$
Since $F_0 < F_e$, We infer that there is no significant difference btw rows	Since $F_0 < F_e$, We infer that there is no significant diff. btw Columns	Since $F_0 > F_e$, We infer that there is significant diff. btw Treatments.

6) Write a detailed account on the following

i) Regression Analysis

ii) Time series Analysis

iii) Factor Analysis

iv) Cluster Analysis

v) Discriminant Analysis.

(i) Regression Analysis: It is useful for prediction purpose

* Simple Regression Models

* Multiple Regression Models

Simple Regression Models/Analysis.

Leaf Weight (y) grams	Leaf length (x) cm
5	4.2
7	5.1
10	6.2
12	7.1
15	7.4
20	8.0

Regression Model of y on x

To estimate y for a certain Value of x

$$(y - \bar{y}) = b_{yx} (x - \bar{x})$$

b_{yx} - regression coefficient of y on x

$$b_{yx} = \frac{\frac{\sum xy}{n} - \bar{x}\bar{y}}{\frac{\sum x^2}{n} - (\bar{x})^2}$$

Eg:

Regression equa. of y on x:

$$y = 0.76x + 0.8$$

For a unit measure in leaf length, the weight increases by 0.76 units.

Regression model of x on y

To estimate x for a certain Value of y

$$(x - \bar{x}) = b_{xy} (y - \bar{y})$$

b_{xy} - regression coefficient of x on y

$$b_{xy} = \frac{\frac{\sum xy}{n} - \bar{x}\bar{y}}{\frac{\sum y^2}{n} - (\bar{y})^2}$$

Eg:

Regression equa. of x on y:

$$x = 1.2y - 0.5$$

For a unit measure in leaf Weight the length increases by 1.2 units

Multiple Regression Analysis: (more than 2 Variables)

$$y = f(x_1, x_2, x_3, x_4)$$

y - dependent Variable

x_1, x_2, x_3, x_4 - independent Variables

Example:

$y \Rightarrow$ yield of a crop

Influenced by a host of Variables

- $x_1 \Rightarrow$ Rainfall (cm)
- $x_2 \Rightarrow$ Soil Fertility (ppm)
- $x_3 \Rightarrow$ Level of Technology (T_1, T_2, T_3)
- $x_4 \Rightarrow$ Farmer's Education ($< M, M-G, > G$)

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4$$

$\beta_0, \beta_1, \dots, \beta_4$ are associated Variables of x .

$\beta_0 \Rightarrow$ Effect of contribution of "other factors" are not included in the Model.

$\beta_1 \Rightarrow$ For a unit increase in rainfall, what is the likely increase in yield.

$\beta_2 \Rightarrow$ For a unit increase in soil condition, what is the likely increase in yield.

$\beta_3 \Rightarrow$ For a unit increase in level of technology, what is the likely increase in yield.

$\beta_4 \Rightarrow$ For a unit increase in Farmer's education, what is the likely increase in yield.

Eg:

$$y = 0.40 + 1.6x_1 + 0.9x_2 + 14.5x_3 + 0.06x_4$$

where $\beta_0 = 0.4$; $\beta_1 = 1.6$; $\beta_2 = 0.9$; $\beta_3 = 14.5$; $\beta_4 = 0.06$

(ii) Time Series Analysis:

Series of data relating to time factor arranged in a sequence manner. Time may be Year, month, day, hour. It consists of ups and downs.

Eg:

Temperature Reading in the Month of Sep' 2017

Day	Temperature (°C)
1	38.4
2	40.2
3	33.5
4	38.5
⋮	⋮
30	38.6

Profit (2011-17)

Year	Profit (in crores)
2011	2.1
2012	4.5
2013	6.5
2014	6.8
2015	7.2
2016	6.9
2017	5.8

Components of Time series:

1. Trend
2. Seasonal Variation
3. Cyclical Variation
4. Irregular Variation

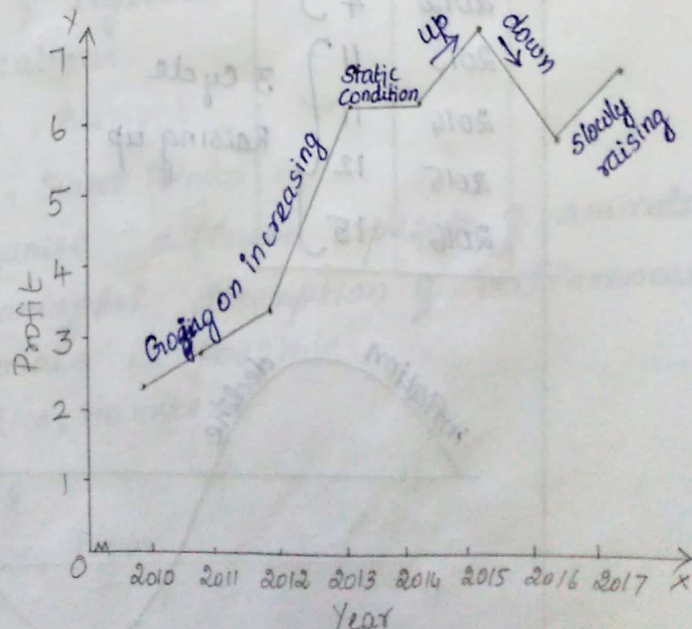
Trend: (Atleast 5 observations)

Time series which have

Increase → Static → Decreasing is called trend.

Profit of Mr. X (in crores)

Year	Profit
2010	2.4
2011	2.8
2012	3.4
2013	6.2
2014	6.2
2015	7.4
2016	5.9
2017	6.6



Seasonal Variation: (Prediction purpose)
Sales, climatic condition)

Sales in Textiles (2013-17)

Year	Quarters			
	Q ₁	Q ₂	Q ₃	Q ₄
2013	14	22	16	10
2014	13	26	17	8
2015	10	24	14	11
2016	12	25	13	10
2017	14	30	15	6

Conclusion: Q₂ → plays a major role.
Q₂ = Festival.

Cyclical Variation: (More than 8 observations)

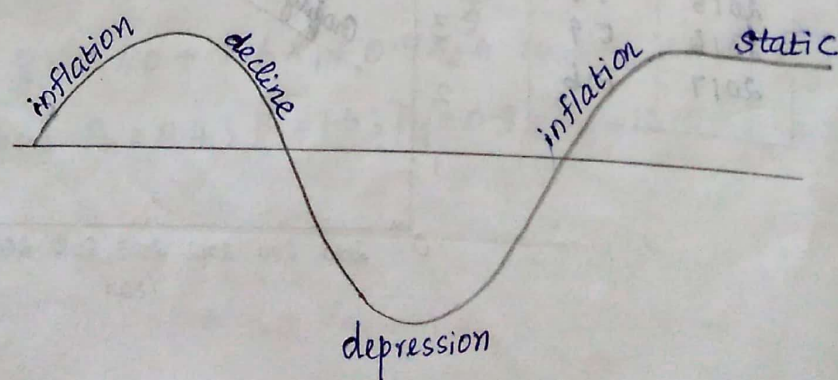
Sales of X & Co (in lakhs)

Year	Sales
2005	4
2006	6
2007	7
2008	8
2009	6
2010	5
2011	4
2012	4
2013	11
2014	11
2015	12
2016	15

1 cycle
Raising up

2 cycle
Coming down

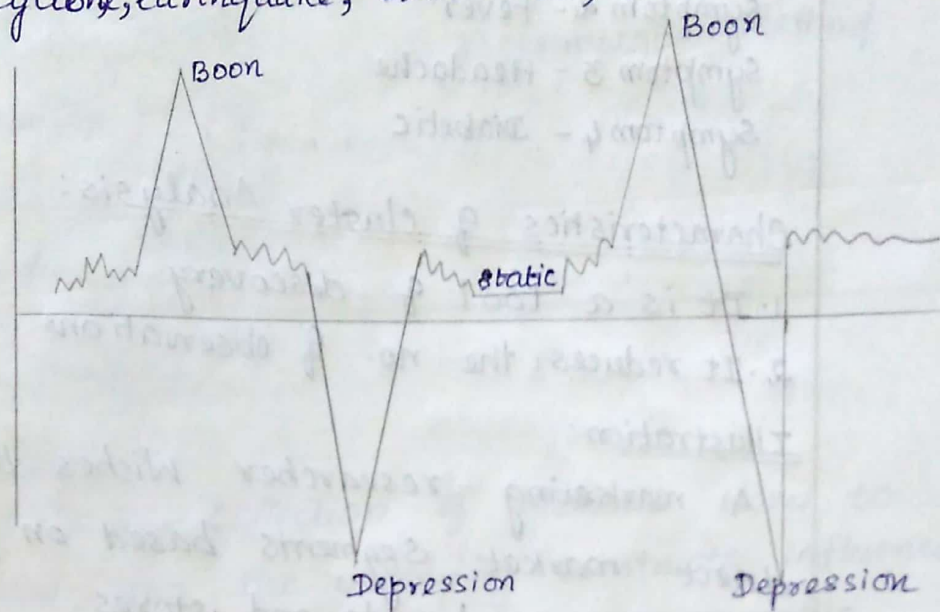
3 cycle
Raising up



Irregular Variation: (Prediction is less precise)

It consists of Boons, depressions at time static. in wide spread and at time static in short interval.

Eg: cyclone, earthquake, Gold rate, etc.



(iv) Cluster Analysis:

Meaning: It is a major technique for classifying "mountain" of information into manageable meaning level. It is a data reduction that creates subgroups and grouping similar objects.

Examples:

1) Super market displays items of similar nature such as types of Vegetables in the same row or nearby locations.

Row 1 - Carrot, Beetroot

Row 2 - Potato, Sweet Potato

2) Biologists organise different species of animals before a meaningful description of differences between animals is possible.

Category 1 - Mice, Hamster

Category 2 - Pig

Category 3 - Lion, Tiger

3) In Medicine, the clustering of symptoms, medicines and diseases leads to taxonomic illness.

Symptom 1 - Cold

Symptom 2 - Fever

Symptom 3 - Headache

Symptom 4 - Diabetic

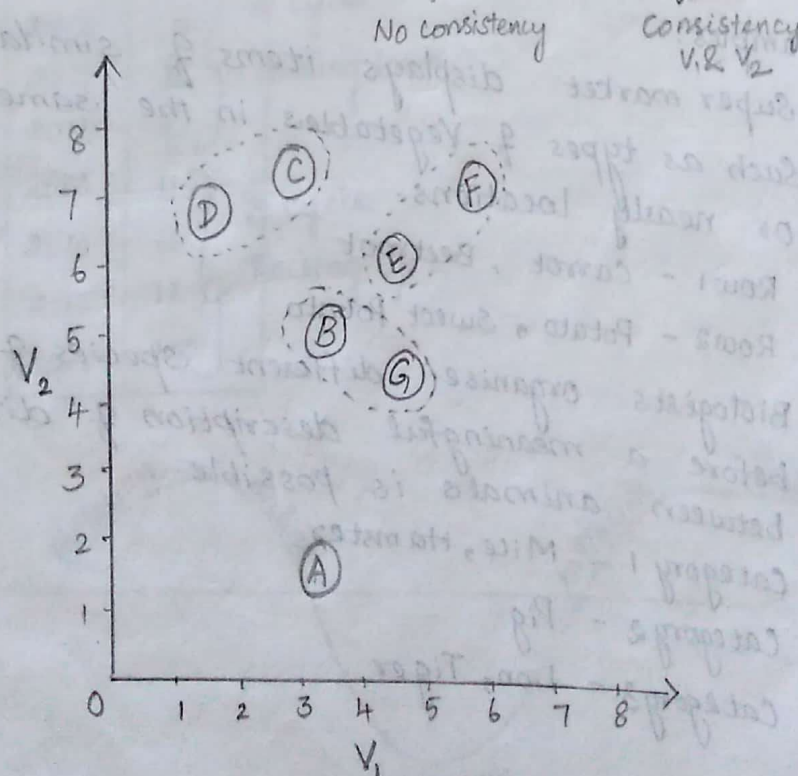
Characteristics of cluster Analysis:

1. It is a tool of discovery
2. It reduces the no. of observations.

Illustration:

A marketing researcher wishes to detect market segments based on patterns of royalty to brands and stores.

Clustering Variable	Respondents						
	A	B	C	D	E	F	G
V_1 : Store Royalty	3	4	4	2	6	7	6
V_2 : Brand Royalty	2	5	7	7	6	7	4



Techniques:

Clustering

Hierarchical clustering
(Based on values)

- 1) simple linkage
- 2) complete linkage
- 3) Centroid Method
- 4) Ward's Method

Non-Hierarchical Cluster

- 1) sequential threshold
- 2) Optimization Method

(iii) Factor Analysis:

Meaning:

It is a collection of Methods used to examine how the underlying constructs influence the responses of a number of measurable Variables.

It is a multivariate data reduction technique. All the Variables under investigation are analysed together to extract underlying Factors. It helps in identifying structure of the data.

Example:

1) Psychology - Basic dimensions of human behaviour

- 1) Memory
- 2) Imagination
- 3) Judgement

2) In Mental testing - specialized abilities namely

- 1) Clerical
- 2) Numerical
- 3) Verbal

Characteristics of FA: FA is keyed to four issues

- 1) Data Reduction/Summarization
- 2) Variable selection
- 3) specifying unit of analysis

Procedure:

- 1) collect Measurement
- 2) Form Correlation Matrix
- 3) Kaiser criterion screen test
- 4) Express initial Variables
- 5) Rotate Factors for a situation
- 6) Interpret Factor Structure.

Illustration:

Final score of a Engineering student depends on the Factors:

x_1 : Rural/Urban

x_2 : Grade in +2

x_3 : Progress in 1-7 semesters

x_4 : Internal component

x_5 : Duration of study in a day

x_6 : Hostler / Day scholar

The corresponding Correlation Matrix,

	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						

(iv) Discriminant Analysis:

Meaning: DA joins a normally scaled criteria or dependent variable with one or more independent variables that are in the interval scale or ratio scale. This is done by computing a discriminant function

$$D_i = d_0 + d_1 x_1 + d_2 x_2 + \dots + d_p x_p$$

where $D_i \Rightarrow$ Score of the discriminant fn.

$d_0 \Rightarrow$ Constant

$d_i's \Rightarrow$ Weighing coefficients.

$x_i's \Rightarrow$ Values of the discriminating Variables used in the analysis.

Example:

Assume We have supervising Ratings that enables administrators as successful or unsuccessful on administrative purposes. There are 3 parameters:

X_1 : Ability to work with others

X_2 : Motivation for administrative work

X_3 : General professional skills

$$D = 0.06X_1 + 0.45X_2 + 0.30X_3$$

We conclude that the ability to work with others is less important than other two variables included in the model.

If he improves X_1 coefficient, he is successful as X_2 & X_3 coefficients are quite high.

Characteristics of Discriminant Analysis:

1. To predict which are the variables relatively discriminates.
2. To classify the groups.

Illustration:

Buying a product Yarn.

X_1 : Durability

X_2 : lightness in weight

X_3 : low investment

X_4 : Rot resistance

Scale 1-10

	X_1	X_2	X_3	X_4	Discriminant score
Buyer 1	3	5	4	3	-1.444
Buyer 2	9	5	6	5	1.574
Buyer 3	8	7	4	4	1.379
Buyer 4	8	6	5	5	1.089
Non buyer 1	3	6	6	3	-1.875
Non buyer 2	4	6	5	2	-0.912
Non buyer 3	7	2	3	2	1.538
Non buyer 4	5	5	6	4	-0.741

$$D_i = d_0 + d_1 x_1 + d_2 x_2 + d_3 x_3 + d_4 x_4$$

$$= -1.8 + 0.618x_1 - 0.055x_2 - 0.188x_3 - 0.157x_4$$

$$D_1 = -1.8 + 0.618(3) - 0.055(5) - 0.188(4) - 0.157(3)$$

$$= -1.444$$

Conclusion:

1. Out of 4 buyers, three buyers classified correctly (D_i 's > 0)
2. Out of 4 non buyers, three buyers classified correctly (D_i 's < 0)

7) Explain the following methods of sampling with Example

- i) Simple Random sampling
- ii) Stratified sampling
- iii) Systematic sampling
- iv) cluster sampling

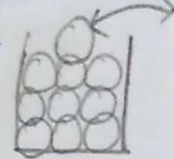
(i) Simple Random Sampling:

Meaning:

- i) This method is adopted only when population is homogeneous.
- ii) Let the size of the population be 'N'.
- iii) After careful thought the sample size has been fixed as 'n'.
- iv) Select 'n' units out of 'N' units randomly.

Procedure For with Replacement: (Equal prob. within selection)

- i) The probability of selecting first unit = $\frac{1}{N}$
- ii) The probability of selecting 2nd unit = $\frac{1}{N}$
- iii) The probability of selecting nth unit = $\frac{1}{N}$



Procedure for Without Replacement: (Unequal prob. within selection)

- i) The probability of selecting 1st unit = $\frac{1}{N}$
- ii) The probability of selecting 2nd unit = $\frac{1}{N-1}$
- iii) The probability of selecting nth unit = $\frac{1}{N-(n-1)}$

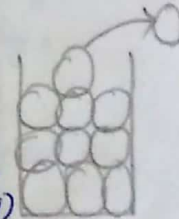


Illustration: There are 10 apples in a basket select 3 apples randomly. (i.e) $N=10$; $n=3$

With Replacement	Without Replacement
$P_1 = \frac{1}{10}$	$P_1 = \frac{1}{10}$
$P_2 = \frac{1}{10}$	$P_2 = \frac{1}{9}$
$P_3 = \frac{1}{10}$	$P_3 = \frac{1}{8}$

(ii) stratified sampling:

Meaning: This method is adopted when population is Heterogeneous. Stratification means classification based on characteristics.
Stratum - singular (homogeneous)
Strata - groups (heterogeneous)

Illustration:

Stratification - Region

Region	No. of persons
Southern	250 = N_1
Northern	500 = N_2
Eastern	100 = N_3
Western	150 = N_4
	$N=1000$

Select 100 persons from 1000.

$$\text{Here } N = N_1 + N_2 + N_3 + N_4$$

$$N = 1000$$

$$n = 100$$

Proportional Allocation: $n_i = \frac{N_i}{N} \times n$

$$n_1 = \frac{250}{1000} \times 100 = 25$$

Select 25 persons out of 250 randomly from Southern region

$$n_2 = \frac{500}{1000} \times 100 = 50$$

Select 50 persons out of 500 randomly from Northern region

$$n_3 = \frac{100}{1000} \times 100 = 10$$

Select 10 persons out of 100 randomly from Eastern region

$$n_4 = \frac{150}{1000} \times 100 = 15$$

Select 15 persons out of 150 randomly from Western region

Note: The more the stratum size, the more the sample size.

Structure to follow: (Computational Layout)

Stratum	Stratum Size	No. of samples	Remarks
1	N_1	$n_1 = \frac{N_1}{N} n$	Proportional allocation Method $n_i = \frac{N_i}{N} n$
2	N_2	$n_2 = \frac{N_2}{N} n$	
$\vdots$			
h	N_h	$n_h = \frac{N_h}{N} n$	

Selecting sample by similar procedures is known as Stratified Random Sampling. (STRS)

(iii) Systematic Sampling Method:

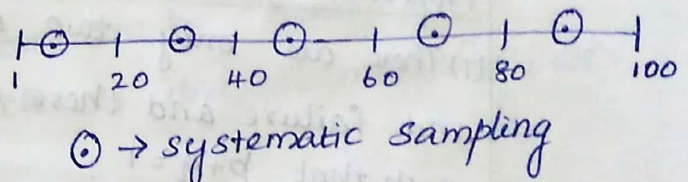
This method is adopted only when population is homogeneous.

$N = nk$ where N - Population size
 n - sample size
 k - systematic sample number
 n, k - must be integers.

Let $N = 100$ students

$n = 5$ students

$$\therefore k = \frac{N}{n} = \frac{100}{5} = 20$$



Select a number from 1 to 20 randomly,

Let the number be 9.

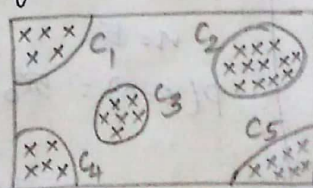
The other samples are automatically selected.

9 29 49 69 89

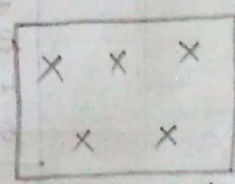
(iv) Cluster Sampling Method: (Homogeneous)



Uniform
(Never this situation)



Concentrated
(units within clusters are similar)



Scattered

- 8) Write a detailed account of the following
- Binomial distribution
 - Poisson distribution
 - Geometric distribution
 - Normal distribution
 - Exponential distribution
 - Weibull Distribution

(i) Binomial distribution

A random Variable x is said to follow Binomial distribution if it assumes non-negative values and its probability mass function is given by

$$P(X=x) = p(x) = {}^nC_x p^x q^{n-x}, \quad x=0,1,2,\dots$$

$q=1-p$

Here p - denotes probability of success

q - denotes probability of failure.

Mean of Binomial distribution $= np$

Variance of Binomial distribution $= npq$

Properties of Binomial distribution:

(1) There are only two outcomes, namely success and failure and these are denoted by p and q such that $p+q=1$

(2) Each trial is independent of others.

(3) The number of trials is finite namely n .

Eg:

x	f
0	50
1	20
2	15
3	10
4	3
5	2
$N=100$	

x : No. of seeds germinated

f : No. of pots

Here $x=0,1,2,3,4,5$

$n=5$

$$P(X=0) = {}^5C_0 p^0 q^{5-0}$$

Applications of Binomial distribution:

(1) Statistical Quality Control

(2) Plant studies

(3) Military Application (Hit/Miss)

(ii) Poisson distribution:

A random variable x is said to follow Poisson distribution if it assumes non-negative values and its probability mass function is given by

$$P(x=x) = p(x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!}, & x=0,1,2,\dots \\ & \lambda > 0 \\ 0 & , \text{otherwise} \end{cases}$$

Mean of Poisson distribution = λ

Variance of Poisson distribution = λ

Properties of Poisson distribution:

(1) It is the limiting case of Binomial distribution under the following conditions, binomial distribution may lead to Poisson distribution

- Let n be the no. of trials, $n \rightarrow \infty$
- Let the probability of success be p , $p \rightarrow 0$
- np is finite.

(2) Poisson distribution can be used for rare events such as deaths, mistakes, etc.

Ex:

Distribution of occurrence of Flood in 1900-2000

No. of floods occurred x	No. of years f
0	89
1	5
2	3
3	2
4 $\geq$	1
$N=100$	

$x=0,1,2,\dots$

Occurrence of Printing Mistakes in a Dictionary

No. of Printing Mistakes x	No. of pages f
0	900
1	60
2	20
3	15
4	3
5 $\geq$	2
$N=1000$	

Applications of Poisson distribution:

- (1) Quality Control
- (2) Quality Assurance
- (3) Reliability Analysis.

(iii) Geometric distribution:

(Regular event but limit not specified)

A random Variable x is said to have a Geometric distribution if it assumes only non-negative values and its probability Mass function is given by

$$P(X=x) = p(x) = \begin{cases} q^x p & , x = 0, 1, 2, \dots, 0 \leq p \leq 1 \\ 0 & , \text{Otherwise} \end{cases}$$

Eg:

Busy Time x	No. of f inefficiencies
9:00	0
9:05	5
9:10	7
9:15	3
9:20	0
9:25	2
9:30	4
9:35	0
9:40	3
9:45	2

Applications of Geometric distribution:

(1) Finding inefficiency of a telephone exchange system during busy period.

(2) Helping Managers to reduce system trials occurring prior to success to reduce cost.

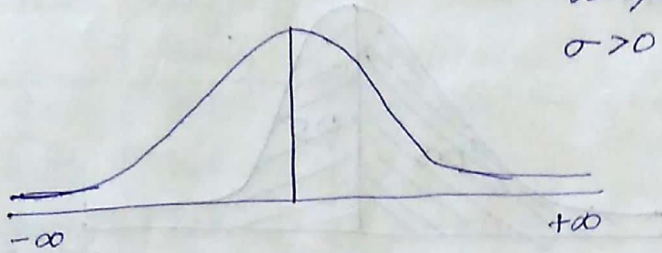
(iv) Normal distribution:

A random Variable X is said to follow Normal distribution with parameters μ (called Mean) and σ^2 (called Variance) if its probability density function is given by the probability law.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$$

$$-\infty < \mu < \infty$$

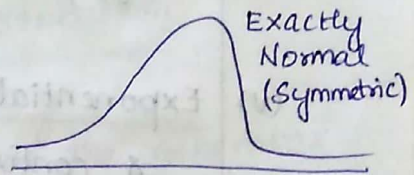
$$\sigma > 0$$



Eg:

Leaf length (in cm) x	Samples f No. of leaves				
	A	B	C	D	E
2.5 - 4.5	20	20	20	20	10
4.5 - 6.5	30	25	22	70	15
6.5 - 8.5	50	32	34	30	25
8.5 - 10.5	30	30	29	15	32
10.5 - 12.5	20	18	17	5	11

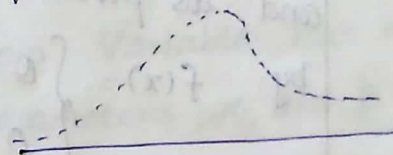
Graph of A:
Ideal situation
Which never occurs
Practically



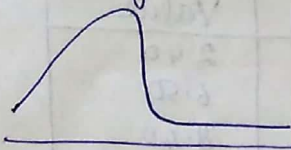
Graph of B:
Approximately Normal



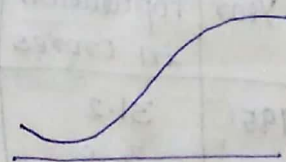
Graph of C:
Approximately Normal



Graph of D:
Asymmetrical



Graph of E:
Asymmetrical

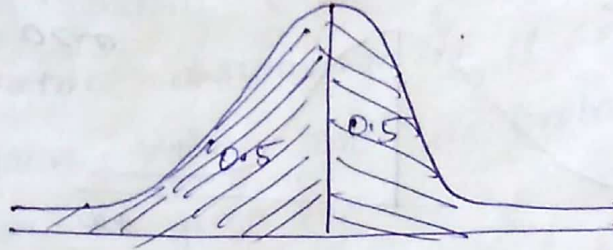


Properties of Normal distribution:

- (1) The curve is BELL shaped and symmetrical
- (2) The range of x is $-\infty < x < \infty$
- (3) The curve is Asymptotic to x axis
(interval take the x -values)
- (4) Coefficient of skewness $\beta_1 = 0$, coefficient of kurtosis $\beta_3 = 3$

(5) $QD:MD:SD = 10:12:15$

(6) Total area under normal curve



Never we will get 100% Truth in any experiment

True Not true / False / Error

95% 5%

99% 1%

66% 32%

(v) Exponential distribution:

A continuous random Variable x assuming non-negative Values is said to have exponential distribution with parameter θ and its probability density function is given

$$f(x) = \begin{cases} \theta e^{-\theta x} & , x \geq 0 \\ 0 & , \text{otherwise} \end{cases}$$

Eg: Increasing Exponential

Year	Population in crores
1951	31.2
1961	34.8
1971	36.8
1981	42.8
1991	44.5
2001	51.6
2011	62.8

Market share Value	
Month	Value
1	2.40
2	6.50
3	8.40
4	9.60
5	9.62
6	11.2
7	14.8
8	22.6
9	35.8
10	36.9
11	44.1
12	62.5

Eg: Double Exponential

For Queuing Theory

Time	Length of Queue
9:00	100
9:30	140
10:00	260
10:30	540
11:00	1100
11:30	300
12:00	700
12:30	2000
1:00	2400

increasing

Break

increasing

Applications of Exponential distribution:

(1) Reliability and Queuing Theory

(2) To find the probability that the generator will break down in the next 20 days.

(vi) Weibull distribution:

A Continuous random Variable has a Weibull distribution with parameters α and β if its density function is given by

$$f(x) = \begin{cases} \alpha \beta x^{\beta-1} e^{-\alpha x^\beta} & , x > 0, \alpha > 0, \beta > 0 \\ 0 & , \text{Otherwise} \end{cases}$$

Weibull distribution is a kind of Mixed distribution based on exponential $e^{-\alpha x^\beta}$.

9) Write a detailed account of the following
 i) t-test ii) F-test iii) χ^2 test

i) Student's t-distribution: (t-test)
 (To study sample characteristics)
 If $x_1, x_2, \dots, x_n$ is a sample of size n
 from a normal population with unknown μ
 and unknown Variance ' σ^2 ', and if $\bar{x} = \frac{\sum x}{n}$
 and $s^2 = \frac{\sum (x_i - \bar{x})^2}{n}$ then

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

$$x_i \quad x_i - \bar{x} \quad (x_i - \bar{x})^2$$

$$x_1 \quad x_1 - \bar{x} \quad (x_1 - \bar{x})^2$$

$$x_2 \quad x_2 - \bar{x} \quad (x_2 - \bar{x})^2$$

$$x_3 \quad x_3 - \bar{x} \quad (x_3 - \bar{x})^2$$

$$\vdots \quad \vdots \quad \vdots$$

$$x_n \quad x_n - \bar{x} \quad (x_n - \bar{x})^2$$

$$\sum x_i \quad 0 \quad \sum (x_i - \bar{x})^2$$

Applications of t-distribution:

(1) Test for sample Mean

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

Note: If μ is given then we can use the same
 If μ is not given then $\mu = 0$

(2) Test for the difference between two sample Means:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{(n_1+n_2-2)} \quad \text{where } s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

Where n_1, n_2 - size of 1st and 2nd sample respectively
 $\bar{x}_1, \bar{x}_2$ - Mean of I and II sample respectively

(3) Paired t-test:
(Studying two contrasting situation)

$$t = \frac{\bar{d}}{\sqrt{\frac{s_d^2}{n}}}, \quad s_d^2 = \frac{1}{n-1} \left[\sum d_i^2 - \frac{(\sum d_i)^2}{n} \right]$$

$$\bar{d} = \frac{\sum d_i}{n}$$

(4) Test for correlation coefficient:

$$t = \frac{r}{\sqrt{1-r^2}} \sqrt{n-2} \sim t_{n-2}$$

Eg: Following is record of scores obtained by 5 students before and after a training. Test whether the training has any impact on the performance of students at 5% level of significance.

Students	1	2	3	4	5
Before training	10	9	10	15	10
After training	12	15	9	18	10

Sol:

i) Null hypothesis: $H_0: \bar{d} = 0$
Alternate hypothesis: $H_a: \bar{d} \neq 0$

ii) Level of significance, $\alpha = 0.05$

iii) Construction of test statistics: (Paired data)

$$t = \frac{\bar{d}}{\sqrt{\frac{s_d^2}{n}}}, \quad s_d^2 = \frac{1}{n-1} \left[\sum d_i^2 - \frac{(\sum d_i)^2}{n} \right]$$

$$\bar{d} = \frac{\sum d_i}{n}$$

iv) Computation:

Student	Before training x	After training y	d = y - x	d ²
1	10	10	0	0
2	9	15	6	36
3	10	9	-1	1
4	15	18	3	9
5	10	10	0	0
			$\sum d = 10$	$\sum d^2 = 50$

$$\bar{d} = \frac{\sum d_i}{n} = \frac{10}{5} = 2 ; \quad S_d^2 = \frac{1}{n-1} \left[\sum d_i^2 - \frac{(\sum d_i)^2}{n} \right]$$

$$= \frac{1}{4} [50 - 20] = 7.5$$

$$t = \frac{\bar{d}}{\sqrt{\frac{S_d^2}{n}}} = \frac{2}{\sqrt{1.5}} = 1.63$$

$$\boxed{t_o = 1.63}$$

For 4 d.f at 5% level, $\boxed{t_o = 2.132}$

v) Inference:

Since $t_o < t_e$, there is no impact on the performance of the students by the training.

ii) F-distribution: (F-test)

(To test the equality of Variances)

Let us have two independent samples of sizes n_1 & n_2 . We want to test the null hypothesis $H_0: \sigma_1^2 = \sigma_2^2$.

$$\boxed{F = \frac{S_1^2}{S_2^2}} , \quad S_1^2 > S_2^2$$

$$\text{where } S_1^2 = \frac{\sum (x_i - \bar{x}_1)^2}{n_1 - 1} ; \quad S_2^2 = \frac{\sum (x_i - \bar{x}_2)^2}{n_2 - 1}$$

Eg: Can the following two samples be regarded as coming from the same population?

Sample I	2	4	6	8	10	$n_1 = 5$
Sample II	3	5	7	9		$n_2 = 4$

Sol:

i) Null hypothesis: $H_0: \sigma_1^2 = \sigma_2^2$

Alternate hypothesis: $H_a: \sigma_1^2 \neq \sigma_2^2$

ii) Level of significance, $\alpha = 0.05$

iii) Construction of test statistics:

$$F = \frac{S_1^2}{S_2^2} , \quad S_1^2 > S_2^2 , \quad S_1^2 = \frac{\sum (x_i - \bar{x}_1)^2}{n_1 - 1} ; \quad S_2^2 = \frac{\sum (x_i - \bar{x}_2)^2}{n_2 - 1}$$

iv) Computation:

x_i	$x_i - \bar{x}_1$	$(x_i - \bar{x}_1)^2$	x_i	$x_i - \bar{x}_2$	$(x_i - \bar{x}_2)^2$
2	-4	16	3	-3	9
4	-2	4	5	-1	1
6	0	0	7	1	1
8	2	4	9	3	9
10	4	16			20
30		40	24		

$$\bar{x}_1 = \frac{\sum x_i}{n_1} = \frac{30}{5} = 6$$

$$\bar{x}_2 = \frac{\sum x_i}{n_2} = \frac{24}{4} = 6$$

$$S_1^2 = \frac{\sum (x_i - \bar{x}_1)^2}{n_1 - 1} = \frac{40}{4} = 10$$

$$S_2^2 = \frac{\sum (x_i - \bar{x}_2)^2}{n_2} = \frac{20}{3} = 6.67$$

$$F_0 = \frac{S_1^2}{S_2^2} = \frac{10}{6.67} = 1.5$$

$$F_c = 9.12$$

v) Inference:

Since $F_0 < F_c$, We infer that there is no significant difference between the Variances of 2 samples.

(iii) Chi-square test: Qualitative Variables
(Applicable for both large and small samples)

- 1) To test for fitness of a distribution
- 2) To test for the association between attributes
- 3) To test the homogeneity of Variances.

Eg: In testing a process for generating random numbers, it is observed that in a run of 1000 the digit x occur with the following frequencies f :

x	0	1	2	3	4	5	6	7	8	9
f	1010	990	960	1020	990	1000	1000	1030	1020	980

Sol:

- i) Null hypothesis: All digits occur with equal frequencies
- ii) Level of significance: $\alpha = 0.05$
- iii) Test statistic: $\chi^2 = \sum \left[\frac{(O_i - E_i)^2}{E_i} \right]$ ($\sum O_i = \sum E_i$)
- iv) Computation:

x	O	E	$O_i - E_i$	$(O_i - E_i)^2$	$(O_i - E_i)^2 / E_i$
0	1010	1000	10	100	0.1
1	990	1000	-10	100	0.1
2	960	1000	-40	1600	1.6
3	1020	1000	20	400	0.4
4	990	1000	-10	100	0.1
5	1000	1000	0	0	0
6	1000	1000	0	0	0
7	1030	1000	30	900	0.9
8	1020	1000	20	400	0.4
9	980	1000	-20	400	0.4
	10000	10000			4

$$\chi^2_o = 4$$

$$\chi^2_e = 16.9$$

v) Inference:

Since $\chi^2_o < \chi^2_e$, We infer that all digits occur with equal frequencies.